Real-Time Probabilistic Delineation of Flood Warning Polygons to Reduce Flood Alert Fatigue

INTRODUCTION

CLIMATE CHANGE INCREASES FLOOD INTENSITY

An increasing number of people are affected by floods each year. From 2000-2015, the affected population rose from 58-85 million to 255–290 million people (Tellman et al., 2021). Additionally, flood coverage is increasing at a rate greater than that of the growth in human population. The world population increased by 18.6% from 2000 to 2015, while areas of observed inundation increased by 34.1% (Tellman et al., 2021).

CASE STUDY: HURRICANE HARVEY (2017)

Precipitation around the 800 mm contour was increased by 18.8% due to climate change. By contrast, the total rainfall increased by around 7% (Risser & Wesser, 2017).

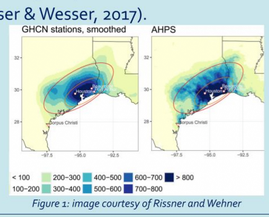


Figure 1: image courtesy of Rissner and Wehner

DAMAGES

In the United States alone, floods annually cause between $179.8 and $496.0 billion in structural damage, reduced tax revenue, commercial hindrance, and infrastructure rebuilding costs (Joint Economic Committee, 2024).

CRY WOLF EFFECT

As climate change makes flooding more frequent and intense, governments are investing in alert systems. However, practices such as overalerting result in false alarms that erode public trust. This erosion of public trust has been coined the cry wolf effect.

CASE STUDY: CENTRAL TEXAS FLOODS (2025)

The 2025 Central Texas Floods killed at least 134 people, the majority of whom disregarded warnings (Rascoe, 2025). In Harris County, one of the affected regions, only 44,000 out of approximately 5 million residents opted for the alert system. Many residents reportedly criticized the county's flooding alert system due to practices like over-alerting, false alarms, and issuing warnings for minor incidents that are not life threatening, causing many residents to dismiss flood warnings as trivial and something to ignore (Vahedifard et al., 2025).

RESEARCH OBJECTIVES

With the rising intensity and coverage of floods, especially in vulnerable areas, a reliable flood prediction model is needed to enable residents to execute measures to protect life and property. In more developed countries such as the U.S., this project aims to improve on pre-existing flood forecasting systems by having higher precision, ideally only warning truly flooded areas to reduce the cry wolf effect. The model should be statistically validated to ensure its usefulness in real-world floods outside of the dataset. To achieve these aims, model performance and statistical standards were used.

As high precision flood predictions serve both developing and developed countries, this was the most important standard to adhere to. The goal was for the model's flood warning polygons to correspond to a real flood 95% of the time. The ground truth flood extent was drawn from the FLASH Flood Observation Database, a public dataset. To validate the efficacy of this model, 33% of the dataset was held out as a test set. This helped ensure that the model generalized to unseen data rather than overfitting to the training data.

BACKGROUND

Most prior work in applying machine learning to flood prediction is narrow in both scope and method. For example, Jang et al. trained a Random Forest model for one urban district in Busan, South Korea, using features tied to simulated outputs and rainfall statistics to predict flood depth. Similarly, Nearing et al. trained a model that predicted flooding in ungauged watersheds worldwide using long short-term memory models and public flood data. While a reliable, valuable model, and in larger scale than the one designed by Jang et al., coastal, tropical storm, snowmelt, and storm flooding are unaccounted for, a gap which our work seeks to address.

Our dual-head ResUNet model, which learns directly from multimodal spatial and temporal inputs, predicts both flood extent and duration. Instead of approximating only a single city's dynamics or those of rivers, our model captures fine-grained flood behavior and produces precise warning polygons across diverse climates, terrains, and geographic features in both urban and rural settings across the world.

DATA HYGIENE & PREPROCESSING

For each flooding event used to construct the dataset, samples of four different categories were involved:

1. ≥20% of pixels are permanent bodies of water
2. at least 7 pixels have a nonzero flood duration (given that they are not permanent water)
3. dry land (i.e., no floods or permanent water)
4. unclassifiable (i.e., everything else at the end)

A total of 40,000 examples were selected for each of the four climate types, yielding 160,000 examples. Examples containing null values in any feature or target variable were discarded. From the remaining pool, 48,486 examples were sampled to create a refined dataset with a 3:1 dry-to-flood ratio per event. The refined dataset was then subjected to an event-level 50-17-33 split to produce training, validation, and testing sets, respectively, ensuring that events used for training were not included in validation or testing and that all sets remained representative of diverse climates, terrains, and special cases.

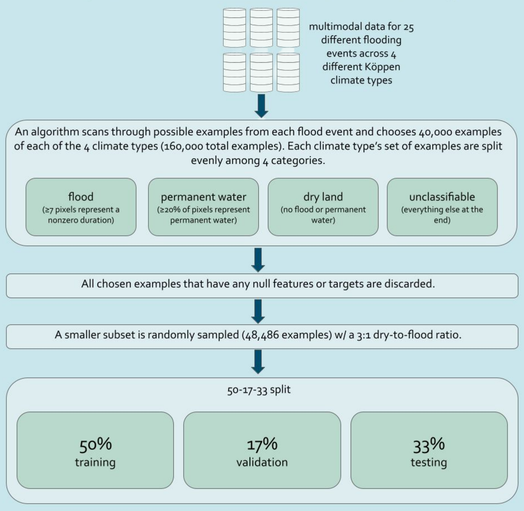
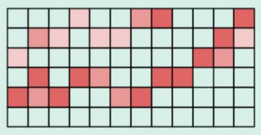


Figure 2: workflow of the data preprocessing steps (created by authors in 2026)

DATASET STRUCTURE & CURATION

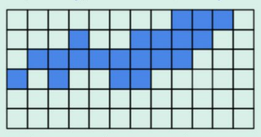
- Flooding can be attributed to a variety of factors including intense rainfall, a lack of vegetation, river overflow, and snowmelt.
- A dataset that embodies the terrain and weather conditions of past examples of flooding was key in developing the ResUNet model.
- Each example in the dataset curated for model training and testing consisted of 17 features and 1 target.
- **Target:**
  - The target was a 64 pixel x 64 pixel raster, where each pixel represented a 250m x 250m plot of land. Each pixel's value was the number of days that the plot of land it represented was flooded during a particular flood event. This raster was procured from a TIF file from the Global Flood Database.

Figure 3: visualizing the flood duration raster, where a darker color indicates more days flooded (created by authors in 2026)



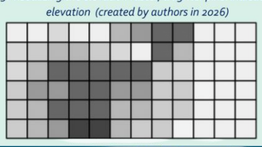
- **Features:**
  - a 64 pixel x 64 pixel raster whose pixels were set to 1 if they represented a plot where a permanent body of water (e.g., a river or lake) exists and 0 otherwise (from a Global Flood Database TIF file for the flood event being studied)

Figure 4: visualizing the permanent bodies of water raster, where a blue pixel indicates the presence of permanent water (created by authors in 2026)



  - a 64 pixel x 64 pixel raster whose pixels represent how many meters above sea level the corresponding plot is (from OpenTopography)

Figure 5: visualizing the elevation raster, a lighter pixel indicates higher elevation (created by authors in 2026)



  - 15 time series data representing hydrometeorological features in the 5 days leading up to noontime on the day of the flood event (from ERA5-Land)
    - The time series features included total precipitation, surface runoff, subsurface runoff, soil water L1, soil water L2, soil water L3, 2m temperature, skin temperature, soil temperature L1, snowmelt, snowfall, evaporation, potential evaporation, U wind 10m, and V wind 10m
- **Visualizing the features and target of an example from the 3 May 2008 Myanmar flooding incident:**

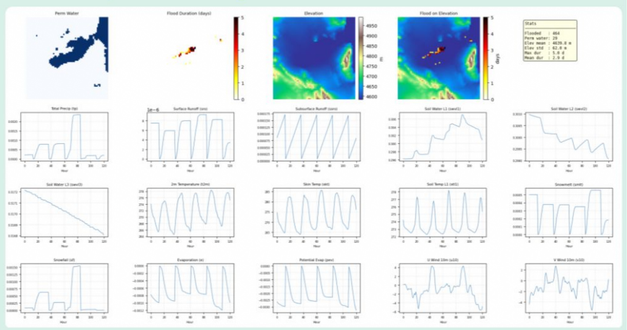


Figure 6: visualizing the features and target of an example from the 3 May 2008 Myanmar flooding incident (created by authors in 2026)

The examples in the training and testing datasets were sampled from an assortment of 25 different flooding incidents at different points in time in various countries. Some of these incidents included a heavy raining event on 18 July 2015 in the southern United States, another heavy raining event on 16 April 2010 in Indonesia's Banjarmasin region, and inundations across the United Kingdom on 05 December 2015.

- **More Information on Databases:**
  - The Global Flood Database is maintained by Google Earth Outreach and the Dartmouth Flood Observatory. It records maps of flooding events across the globe.
  - OpenTopography hosts Copernicus DEM 30m dataset.
  - ERA5-Land, maintained by the European Union, provides a host of hydrometerological time series for all regions of the globe.

ENGINEERING METHODOLOGY

We developed a dual-head Residual U-Net (ResUNet) that converts multimodal geospatial inputs into probabilistic flood warning polygons. Static inputs provide geographic information, while a temporal encoder summarizes weather conditions. These features are fused and processed by an encoder–decoder that predicts both per-pixel flood probability and flood duration, which are then converted into GeoJSON warning polygons.

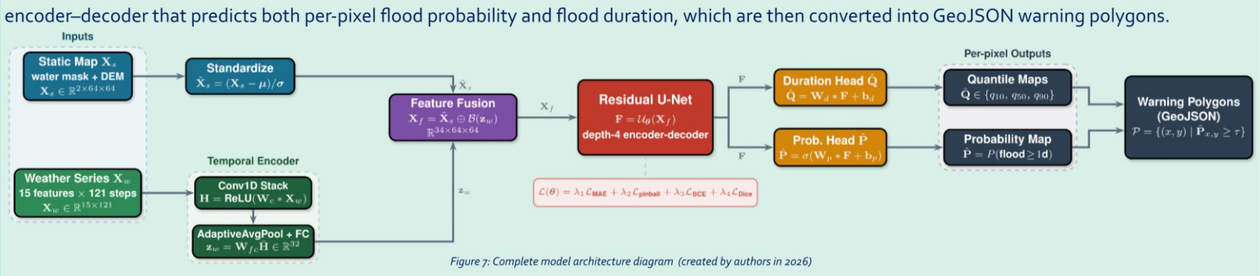


Figure 7: Complete model architecture diagram (created by authors in 2026)

1. Multimodal Inputs

Static spatial rasters encode location-specific geographic vulnerability, while the **weather time series** captures storm evolution over time. Together, these inputs allow the model to learn both *where* flooding is physically likely and *when/how* hydrometeorological conditions make flooding probable.

$$\mathbf{X}_s \in \mathbb{R}^{2\times64\times64}, \qquad \mathbf{X}_w \in \mathbb{R}^{15\times121} \tag{1}$$

where $\mathbf{X}_s$ contains the permanent water mask and digital elevation model (DEM), and $\mathbf{X}_w$ contains 15 hydrometeorological variables across 121 time steps.

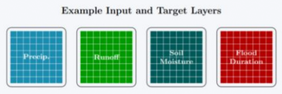


Figure 2. Example input layers used by the model (created by authors in 2026).

5. Dual Output Heads

The shared ResUNet feature map branches into two heads. One head predicts **per-pixel duration quantiles**, while the other predicts **per-pixel flood probability**.

$$\hat{\mathbf{Q}} = W_d * \mathbf{F} + b_d \qquad \hat{\mathbf{P}} = \sigma(W_p * \mathbf{F} + b_p) \tag{8}$$

$$\hat{\mathbf{Q}}(x,y) = \big(\hat{q}_{10}(x,y),\ \hat{q}_{50}(x,y),\ \hat{q}_{90}(x,y)\big) \tag{9}$$

Output	Interpretation
q_{10}	Conservative lower-bound
q_{50}	Median estimate
q_{90}	Upper-bound estimate
$\hat{P}$	Prob. flooding ≥ 1 day

Table 1. Model outputs (created by authors in 2026).

2. Temporal Encoder

Flooding depends not only on instantaneous rainfall, but also on **temporal patterns**. A one-dimensional convolutional encoder extracts these patterns from the raw weather sequence before fusion with the spatial inputs.

$$\mathbf{H} = \text{ReLU}(W_c * \mathbf{X}_w) \tag{2}$$

$$\mathbf{z}_w = W_{fc}\,\bar{\mathbf{H}} \in \mathbb{R}^{32} \tag{3}$$

Here, W_c denotes the learned Conv1D filters, $\bar{\mathbf{H}}$ is the pooled temporal representation, and $\mathbf{z}_w$ is the final 32-dimensional weather embedding.

6. Polygon Generation

The probability map is converted into a warning product by thresholding predicted risk at a selected confidence level τ.

$$\mathcal{P} = \{(x,y) \mid \hat{P}_{x,y} \geq \tau\} \tag{10}$$

This transforms the network output into a georeferenced flood-warning polygon.

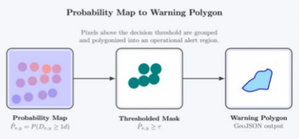


Figure 5. Pipeline to final warning polygon (created by authors in 2026).

3. Feature Fusion

The learned weather embedding is **broadcast** across the spatial grid. This weather representation is then concatenated with the standardized static rasters.

$$\tilde{\mathbf{X}}_s = \frac{\mathbf{X}_s - \mu}{\sigma} \tag{4}$$

$$\mathbf{X}_f = \tilde{\mathbf{X}}_s \oplus \mathcal{B}(\mathbf{z}_w) \tag{5}$$

$$\mathbf{X}_f \in \mathbb{R}^{34\times64\times64} \tag{6}$$

where $\mathcal{B}(\mathbf{z}_w)$ denotes broadcasting the weather embedding, and $\oplus$ denotes channel-wise concatenation.

7. Training Objective

The model is trained with a composite loss:

$$\mathcal{L}(\theta) = \lambda_1\mathcal{L}_{\text{MAE}} + \lambda_2\mathcal{L}_{\text{pinball}} + \lambda_3\mathcal{L}_{\text{BCE}} + \lambda_4\mathcal{L}_{\text{Dice}} \tag{11}$$

Loss Term	Role in Training
$\mathcal{L}_{\text{MAE}}$	Numerical accuracy of durations
$\mathcal{L}_{\text{pinball}}$	Trains quantile outputs
$\mathcal{L}_{\text{BCE}}$	Probability estimation
$\mathcal{L}_{\text{Dice}}$	Spatial overlap quality

Table 2. Training objective components (created by authors in 2026).

4. ResUNet Backbone

Flood delineation is a **dense spatial prediction** task. We therefore use a **Residual U-Net**, whose encoder–decoder structure captures both large-scale terrain–storm context and fine local detail.

$$\mathbf{F} = U_\theta(\mathbf{X}_f) \tag{7}$$

The **encoder** downsamples the fused tensor, while the **decoder** upsamples the representation. **Skip connections** preserve high-resolution spatial detail.

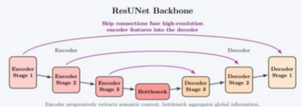


Figure 4. Conceptual ResUNet encoder–decoder (created by authors in 2026).

8. Why This Architecture?

Design Choice	Why It Matters
Conv1D Encoder	Captures weather persistence/lag effects
Feature Fusion	Combines storm context with local geography
ResUNet	Preserves sharp boundaries in spatial prediction
Dual-head	Predicts likelihood and duration uncertainty
Polygonization	Creates operational products

Table 3. Engineering rationale (created by authors in 2026).

RESULTS

The dual-head Residual U-Net successfully predicted both flood extent and flood duration across held-out test samples. Compared to the average National Weather Service flood warning polygon, the model reduced false warning area by **38.2%** while preserving **99.4%** of true flooded area. Additionally, the model predicted per-pixel flood duration with median estimates averaging approximately 1.5 days from the observed values. On top of generating a final warning polygon, the system produced per-pixel flood probabilities and duration quantiles, providing a richer and more uncertainty-aware representation of flood risk. These results provide more spatially precise flood warnings and reduced overwarning that will increase public trust in evacuation alerts.

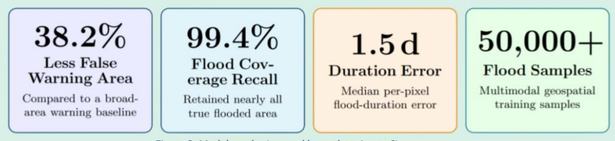


Figure 8: Model results (created by authors in 2026)

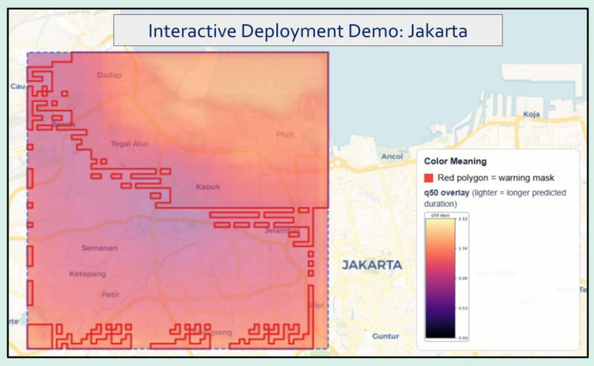


Figure 9: Example model output on Jakarta (created by authors in 2026)

To demonstrate practical usability, we also developed a lightweight front-end interface that displays predicted flood probability, flood duration, and warning polygons for selected cities. In flood-prone examples such as Jakarta, the system produces clear risk regions and warning geometry. The system is designed not only to classify flood risk, but to produce a warning output that is operationally interpretable. Rather than issuing a coarse binary alert over an entire region, the model generates a probability field that can be converted into a tighter and more spatially precise warning polygon that produces fewer false warnings.

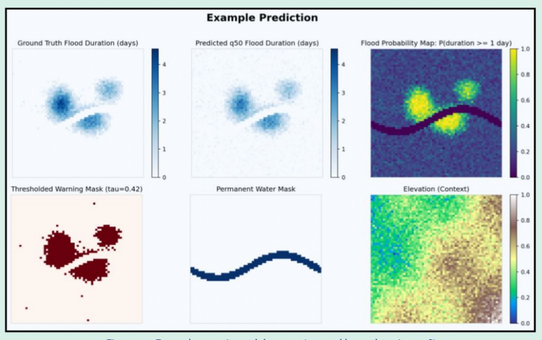


Figure 10: Example generic model output (created by authors in 2026)

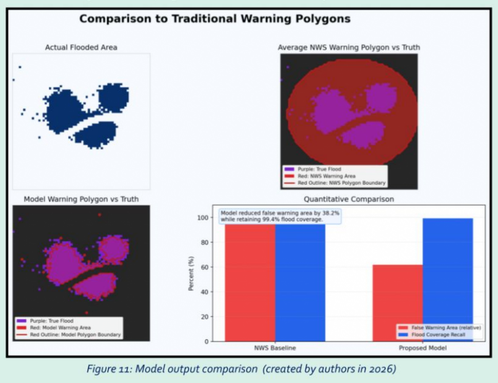


Figure 11: Model output comparison (created by authors in 2026)

Compared to broad-area warning baselines for flood warnings such as the NWS, the proposed model generates substantially tighter warning poly- gons while preserving nearly all true flooded area.

COMBATING BIAS BY PRIORITIZING DATA DIVERSITY

- **Floods occur due to many factors, often localized, across the world.**
  - In the Indonesian capital of Jakarta, the weight of urban development combined with residents' overuse of underground aquifers is causing land to sink, increasing flood risk in the coastal city.
  - Global warming is causing seawater to evaporate more quickly, creating intense, cyclone-like storms that are bringing unusual floods to the typically dry deserts of North Africa and Arabia.



Figure 12: flooding in the Indonesian capital of Jakarta (image courtesy of the South China Morning Post)

1) Embodying a Diverse Variety of Climates

One way to classify regions by their weather patterns is using climate types. The Köppen climate classification system splits the Earth into five different climate types: A is tropical, B is arid, C is temperate, D is continental, and E is polar. The dataset for this project was designed so that climate types A, B, C, and D each held ~25% of the examples. Climate type E was excluded because data on floods in such regions are relatively limited, and these regions (such as Antarctica and the North Pole) are not as densely populated as those of the other climate zones.

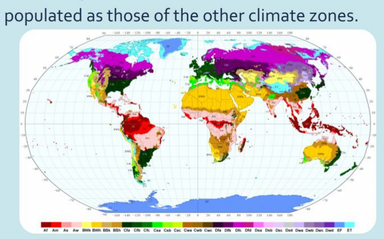


Figure 13: Köppen climate classification (image courtesy of NOAA)

2) Embodying a Diverse Variety of Terrains

A variety of flooding incidents in different countries were chosen as data sources for each climate type. For instance, examples representative of climate type A (tropical) came from flooding events in Myanmar, Mexico, Bangladesh, and Indonesia, countries that each have their own unique topography.

CONCLUSION

As floods become more intense and widespread, the current practice of overwarning regions will reinforce the cry wolf effect. This model is inexpensive and easy to use, enabling use by governments all across the world to more precisely predict flood coverage. Floods in the U.S. alone inflict $125 billion annually, according to the Joint Economic Committee. Therefore, prompt responses, which will reduce damage to property and loss of life, could potentially save billions of dollars worldwide. While this model is a large advance towards precise flood polygon delineation, improvements can still be made.

Currently, the model does not measure lead time, the amount of time between the flood warning and the occurrence of the flood. This variable is key to an effective evacuation response, as a trusted warning may not be able to be acted upon with too little time. Additionally, integration of this model into national governments' services would increase its accessibility. This model may particularly benefit flood-prone countries with underdeveloped weather warning systems, such as Pakistan, India, and Nigeria. Early warnings will allow emergency services and citizens to mitigate fatalities and property damage.

APPLICATIONS & FUTURE WORK

- Ordinary families and town officials might not have the expertise to code their own pipeline for fetching data and running inference. Thus, a web interface can be developed whereby a user inputs a bounding region and simply waits for a map displaying predicted flood zones to appear on screen. The steps of fetching live data, preparing it, and inputting it to the ResUNet model are done automatically.

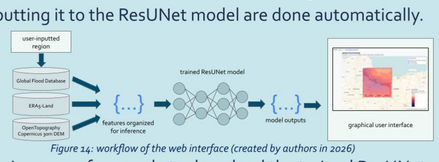


Figure 14: workflow of the web interface (created by authors in 2026)

- Developing a way for people to download the trained ResUNet as a base model and then fine tune it on their own regional data will make the model more versatile.
- To create an improved iteration of the model that involves higher-resolution hydrometeorological and topography data, partnering with local communities that are willing to build infrastructure for data collection will be key.

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